

EXTENDED PHYSICS EXAM NOTES

Serway & Jewett — Physics for Scientists & Engineers, 9th Edition

Units 1-4 | Kinematics · Dynamics · Electrostatics · Circuits

Comprehensive notes with worked examples, key pitfalls & formula sheets

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How to use these notes: Read the concept explanation first, then study the formulas, then work through the examples. Pay special attention to WATCH OUT warnings — these are the most common sources of exam errors.

UNIT 1 — KINEMATICS & VECTORS

Chapters 1, 2, 3, 4 | The language of motion: describing how things move without asking why

Chapter 1 — Physics & Measurement

1.1 SI Base Units

All physical quantities are expressed in terms of seven base SI units. Every other unit is derived from these.

Quantity	SI Unit	Symbol
Length	metre	m
Mass	kilogram	kg
Time	second	s
Electric current	ampere	A
Temperature	kelvin	K
Amount of substance	mole	mol
Luminous intensity	candela	cd

1.2 Dimensional Analysis

Dimensions are the physical nature of a quantity — expressed in terms of [L], [M], [T], [I], [θ]. Dimensional analysis checks whether equations are physically consistent.

- Rule: Every term in an equation must have the same dimensions. You cannot add metres to seconds.
- Useful check: if you derive a formula, verify dimensions before plugging in numbers.

Example — check $v = \sqrt{2gh}$:

Left side: $[v] = L/T$

Right side: $\sqrt{([g][h])} = \sqrt{(L/T^2 \times L)} = \sqrt{(L^2/T^2)} = L/T \checkmark$ (dimensions match)

1.3 Significant Figures & Scientific Notation

- Addition/subtraction: answer has fewest decimal places of any input.
- Multiplication/division: answer has fewest sig-figs of any input.
- Exact numbers (e.g. counted objects, defined conversions) have infinite sig-figs.
- Scientific notation: 3.00×10^8 has 3 sig-figs; 300 is ambiguous (could be 1, 2, or 3).

⚠ WATCH OUT: Do NOT round intermediate results — only round the final answer.

1.4 Unit Conversion

Method: multiply by conversion factors = 1

e.g. $60 \text{ mi/h} \times (1609 \text{ m} / 1 \text{ mi}) \times (1 \text{ h} / 3600 \text{ s}) = 26.8 \text{ m/s}$

Write units at every step and cancel them explicitly.

Practice Exercises & Answers

Q1.

A car travels 150 km in 2.5 hours. What is its average speed in m/s?

Answer:

$$\text{Average speed} = 150\,000 \text{ m} / (2.5 \times 3600 \text{ s}) = 150\,000 / 9000 = 16.7 \text{ m/s.}$$

Q2.

Convert 60 miles per hour to metres per second. (1 mile = 1609 m)

Answer:

$$60 \text{ mi/h} \times (1609 \text{ m/mi}) \times (1 \text{ h} / 3600 \text{ s}) = 60 \times 1609 / 3600 = 26.8 \text{ m/s.}$$

Q3.

The kinetic energy of an object is given by $KE = \frac{1}{2}mv^2$. Check that the dimensions are consistent using dimensional analysis.

Answer:

$[KE] = [m][v^2] = \text{kg} \times (\text{m/s})^2 = \text{kg} \cdot \text{m}^2/\text{s}^2 = \text{J}$. This matches the unit of energy (joule), so the formula is dimensionally consistent.

Q4.

A measurement is recorded as 0.00340 m. How many significant figures does it have?

Answer:

3 significant figures. Leading zeros are not significant; the trailing zero after 4 is significant because it comes after the decimal point.

Q5.

Express the speed of light ($3.00 \times 10^8 \text{ m/s}$) in km/h.

Answer:

$$3.00 \times 10^8 \text{ m/s} \times (1 \text{ km}/1000 \text{ m}) \times (3600 \text{ s}/1 \text{ h}) = 1.08 \times 10^9 \text{ km/h.}$$

Chapter 2 — Motion in One Dimension

2.1 Position, Displacement, Distance

- **Position $x(t)$:** coordinate of object on a number line (can be positive, negative, zero).
- **Displacement $\Delta x = x_f - x_i$:** VECTOR quantity — has sign indicating direction.
- **Distance:** total path length — always positive, path-dependent.

Example: A car goes from $x = 2 \text{ m}$ to $x = -3 \text{ m}$.

Displacement: $\Delta x = -3 - 2 = -5 \text{ m}$ (moved 5 m in the negative direction)

Distance: 5 m (same here, but different if path doubled back)

⚠ WATCH OUT: Displacement $\neq$ Distance. Exam questions often try to confuse these.

2.2 Velocity and Speed

Quantity	Definition	Formula	Sign
Average velocity	Displacement / time	$v_{\text{avg}} = \Delta x / \Delta t$	Can be $\pm$
Instantaneous velocity	Limit as $\Delta t \rightarrow 0$	$v = dx/dt$ (slope of x-t graph)	Can be $\pm$
Average speed	Distance / time	$s_{\text{avg}} = d / \Delta t$	Always ≥ 0
Instantaneous speed	instantaneous velocity	$ v $	Always ≥ 0

2.3 Acceleration

- **Average acceleration:** $a_{\text{avg}} = \Delta v / \Delta t = (v_f - v_i) / (t_f - t_i)$
- **Instantaneous acceleration:** $a = dv/dt = d^2x/dt^2$ (slope of v-t graph)
- On a v-t graph: slope = acceleration; area under curve = displacement.

⚠ WATCH OUT: Deceleration means the acceleration vector points OPPOSITE to the velocity vector — it does NOT necessarily mean a is negative. A car slowing down while going left has

positive a.

2.4 Kinematic Equations (Constant Acceleration)

These FOUR equations apply ONLY when acceleration is constant throughout the time interval. Identify known and unknown variables, then pick the equation that contains them.

The Four Kinematic Equations

(1) $v = v_0 + at$ [no x]

(2) $x = x_0 + v_0t + \frac{1}{2}at^2$ [no v]

(3) $v^2 = v_0^2 + 2a(x - x_0)$ [no t]

(4) $x = x_0 + \frac{1}{2}(v_0 + v)t$ [no a]

Given	Find	Use equation
v_0, a, t	v, x	(1) then (2)
v_0, v, a	t, x	(1) for t, then (2) OR (3)
v_0, a, x	v	(3)
v_0, v, t	x, a	(4) for x, (1) for a

2.5 Free Fall

- Special case of constant acceleration: $a = -g = -9.80 \text{ m/s}^2$ (taking upward as positive).
- At the highest point of a trajectory, $v = 0$ instantaneously, but $a = -9.80 \text{ m/s}^2$ (still!).
- Time to rise equals time to fall for symmetric trajectories.

Worked Example — Ball thrown upward at 20 m/s:

Find max height: $v^2 = v_0^2 + 2a\Delta y \rightarrow 0 = (20)^2 + 2(-9.8)\Delta y \rightarrow \Delta y = 400/19.6 = 20.4 \text{ m}$

Time to max height: $v = v_0 + at \rightarrow 0 = 20 - 9.8t \rightarrow t = 2.04 \text{ s}$

Total time in air: $2 \times 2.04 = 4.08 \text{ s}$ (symmetric motion)

✓ **EXAM TIP:** Always set up a sign convention FIRST. Choose + direction and stick to it throughout.

Practice Exercises & Answers

Q1.

A ball is dropped from rest from a height of 45 m. (a) How long does it take to reach the ground? (b) What is its speed just before impact?

Answer:

(a) $y = \frac{1}{2}gt^2 \rightarrow 45 = \frac{1}{2}(9.8)t^2 \rightarrow t^2 = 9.18 \rightarrow t = 3.03 \text{ s}$. (b) $v = gt = 9.8 \times 3.03 = 29.7 \text{ m/s}$ (downward).

Q2.

A train accelerates uniformly from rest to 72 km/h in 20 s. (a) What is the acceleration? (b) How far does it travel in this time?

Answer:

72 km/h = 20 m/s. (a) $a = \Delta v/\Delta t = 20/20 = 1.0 \text{ m/s}^2$. (b) $x = v_0t + \frac{1}{2}at^2 = 0 + \frac{1}{2}(1.0)(20^2) = 200 \text{ m}$.

Q3.

A car moving at 30 m/s brakes and stops in 6 s. What is the braking acceleration, and how far does it travel before stopping?

Answer:

$a = (0 - 30)/6 = -5.0 \text{ m/s}^2$ (deceleration). $x = v_0t + \frac{1}{2}at^2 = 30(6) + \frac{1}{2}(-5)(36) = 180 - 90 = 90 \text{ m}$.
Check: $v^2 = 30^2 + 2(-5)(90) = 900 - 900 = 0$. ✓

Q4.

A stone is thrown upward at 20 m/s from a bridge 25 m above a river. Find the time it takes to hit the water.

Answer:

Take upward as positive, origin at bridge: $y = 20t - 4.9t^2$. Set $y = -25$ (water level): $4.9t^2 - 20t - 25 = 0$. $t = (20 + \sqrt{(400 + 490)})/9.8 = (20 + 29.8)/9.8 = 5.1$ s (take positive root).

Q5.

On a velocity-time graph, what does the area under the curve between $t = 2$ s and $t = 5$ s represent?

Answer:

The area under a v-t graph equals the displacement of the object during that time interval. If v is in m/s and t in s, the area has units of metres.

Chapter 3 — Vectors

3.1 Scalars vs Vectors

- **Scalar:** described by magnitude only. Examples: mass, temperature, speed, time, energy.
- **Vector:** described by magnitude AND direction. Examples: displacement, velocity, force, acceleration, momentum.

3.2 Vector Components

Any vector A can be broken into perpendicular components. This is the foundation of all 2D and 3D physics.

Component Decomposition

$$A_x = A \cos\theta \text{ (horizontal component)}$$

$$A_y = A \sin\theta \text{ (vertical component)}$$

$$A = \sqrt{(A_x)^2 + (A_y)^2} \text{ (magnitude from components)}$$

$$\theta = \arctan(A_y / A_x) \text{ (angle from +x axis — CHECK QUADRANT!)}$$

$$\text{Vector form: } A = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$$

⚠ WATCH OUT: The arctan function always returns a value between -90° and $+90^\circ$. If the vector is in quadrant II or III, add or subtract 180° to get the correct angle. Always draw a sketch to verify.

3.3 Vector Addition

- **Component method:** $R = A + B \rightarrow R_x = A_x + B_x, R_y = A_y + B_y$
- **Graphical method:** place vectors head-to-tail; resultant goes from first tail to last head.
- **Subtraction:** $A - B = A + (-B)$, where $-B$ has same magnitude but opposite direction.

3.4 Dot and Cross Products

DOT PRODUCT (scalar result):

$$A \cdot B = AB \cos\theta$$

$$= A_x B_x + A_y B_y + A_z B_z$$

Zero when vectors are perpendicular

Used for: work, projections

CROSS PRODUCT (vector result):

$$|A \times B| = AB \sin\theta$$

Direction: right-hand rule

Zero when vectors are parallel

Used for: torque, angular momentum

Practice Exercises & Answers

Q1.

Vector A has magnitude 8 N at 30° above the +x axis. Vector B has magnitude 6 N at 120° above the +x axis. Find A + B.

Answer:

$$A_x = 8\cos 30^\circ = 6.93, A_y = 8\sin 30^\circ = 4.0. B_x = 6\cos 120^\circ = -3.0, B_y = 6\sin 120^\circ = 5.20. R_x = 6.93 - 3.0 = 3.93 \text{ N}, R_y = 4.0 + 5.20 = 9.20 \text{ N}. |R| = \sqrt{(3.93^2 + 9.20^2)} = \sqrt{(15.4 + 84.6)} = \sqrt{100} = 10.0 \text{ N}. \theta = \arctan(9.20/3.93) = 66.8^\circ \text{ above } +x \text{ axis}.$$

Q2.

Find the dot product of $A = 3\hat{i} + 4\hat{j}$ and $B = -2\hat{i} + 5\hat{j}$.

Answer:

$$A \cdot B = (3)(-2) + (4)(5) = -6 + 20 = 14. \text{ Since } A \cdot B > 0, \text{ the angle between them is less than } 90^\circ.$$

Q3.

A vector has components $V_x = -5 \text{ m}$ and $V_y = -12 \text{ m}$. Find its magnitude and direction.

Answer:

$$|V| = \sqrt{(25 + 144)} = \sqrt{169} = 13 \text{ m}. \theta = \arctan(12/5) = 67.4^\circ, \text{ but both components are negative} \rightarrow \text{third quadrant. Direction: } 180^\circ + 67.4^\circ = 247.4^\circ \text{ from } +x \text{ axis (or } 67.4^\circ \text{ south of west)}.$$

Q4.

Explain why you cannot simply add the magnitudes of two vectors to find the resultant magnitude.

Answer:

Vectors have direction as well as magnitude. Adding magnitudes only gives the correct result when both vectors point in exactly the same direction ($\theta = 0^\circ$). In general, you must add components separately: $R_x = A_x + B_x$, $R_y = A_y + B_y$, then find $|R| = \sqrt{(R_x^2 + R_y^2)}$. The triangle inequality gives $|A| - |B| \leq |R| \leq |A| + |B|$.

Chapter 4 — Motion in Two Dimensions

4.1 Position, Velocity & Acceleration Vectors

- Position: $\mathbf{r} = x\hat{i} + y\hat{j}$
- Velocity: $\mathbf{v} = d\mathbf{r}/dt = v_x\hat{i} + v_y\hat{j}$ (tangent to path)
- Acceleration: $\mathbf{a} = d\mathbf{v}/dt = a_x\hat{i} + a_y\hat{j}$ (may point in any direction)

Key principle: x and y motions are INDEPENDENT. Acceleration in x affects only v_x and x. Acceleration in y affects only v_y and y.

4.2 Projectile Motion — Full Analysis

A projectile experiences only gravity (no air resistance). Horizontal motion is constant velocity; vertical motion is free fall.

Projectile Motion Equations (launch from origin at angle θ)

Initial components: $v_{0x} = v_0 \cos\theta$, $v_{0y} = v_0 \sin\theta$

Horizontal ($a_x = 0$): $x = v_{0x} t = v_0 \cos\theta \cdot t$

Vertical ($a_y = -g$): $y = v_{0y} t - \frac{1}{2}gt^2 = v_0 \sin\theta \cdot t - \frac{1}{2}gt^2$

Vertical velocity: $v_y = v_0 \sin\theta - gt$

Speed at any time: $|v| = \sqrt{(v_x^2 + v_y^2)}$

Range (flat ground): $R = v_0^2 \sin(2\theta) / g$

Max height: $H = v_0^2 \sin^2\theta / (2g)$

Time of flight: $T = 2v_0 \sin\theta / g$

- Maximum range occurs at $\theta = 45^\circ$. Complementary angles (e.g. 30° & 60°) give the same range.
- At maximum height: $v_y = 0$, but $v_x = v_0 \cos\theta$ (unchanged throughout).

Worked Example — Ball kicked at 25 m/s at 37° above horizontal:

$$v_{0x} = 25 \cos 37^\circ = 19.97 \text{ m/s}, v_{0y} = 25 \sin 37^\circ = 15.04 \text{ m/s}$$

$$\text{Time to max height: } t = v_{0y}/g = 15.04/9.8 = 1.53 \text{ s}$$

$$\text{Max height: } H = v_{0y}^2/(2g) = (15.04)^2/19.6 = 11.5 \text{ m}$$

$$\text{Total time: } T = 2 \times 1.53 = 3.07 \text{ s}$$

$$\text{Range: } R = v_{0x} \times T = 19.97 \times 3.07 = 61.3 \text{ m}$$

4.3 Uniform Circular Motion

- Speed is constant; direction (and therefore velocity) changes continuously.
- **Centripetal acceleration:** $a_c = v^2/r$, directed toward the center.
- **Period:** $T = 2\pi r/v$. **Frequency:** $f = 1/T$. **Angular speed:** $\omega = 2\pi f = v/r$.

⚠ WATCH OUT: There is no 'centrifugal force' pointing outward. That sensation comes from inertia in a rotating frame — it is NOT a real force and should NEVER appear in a free-body diagram.

4.4 Relative Velocity

Relative Velocity Formula

$$\mathbf{v_{p_A}} = \mathbf{v_{p_B}} + \mathbf{v_{B_A}}$$

or: velocity of A relative to ground = velocity of A relative to B + velocity of B relative to ground

Vector addition — draw a triangle or use components

✓ **EXAM TIP:** Draw the reference frames and label which velocity is relative to which. Subscript order matters: v_{AB} means 'A relative to B'.

Practice Exercises & Answers

Q1.

A ball is kicked at 40 m/s at 37° above the horizontal from flat ground. Find (a) the maximum height, (b) the range, and (c) the speed at the highest point.

Answer:

(a) $v_{0y} = 40 \sin 37^\circ = 24.1 \text{ m/s}$. $H = v_{0y}^2/(2g) = (24.1)^2/19.6 = 29.6 \text{ m}$. (b) $T = 2v_{0y}/g = 2(24.1)/9.8 = 4.92 \text{ s}$. $v_{0x} = 40 \cos 37^\circ = 31.9 \text{ m/s}$. $R = v_{0x}T = 31.9 \times 4.92 = 157 \text{ m}$. (c) At the highest point $v_y = 0$, so speed = $v_{0x} = 31.9 \text{ m/s}$.

Q2.

A car rounds a flat circular curve of radius 80 m at 20 m/s. What centripetal acceleration does it experience? What friction force is needed if $m = 1200 \text{ kg}$?

Answer:

$a_c = v^2/r = 400/80 = 5.0 \text{ m/s}^2$. $F = ma_c = 1200 \times 5.0 = 6000 \text{ N}$ (provided by static friction between tyres and road).

Q3.

Plane A flies north at 400 km/h relative to the ground. Plane B flies east at 300 km/h relative to the ground. What is the velocity of A relative to B?

Answer:

$\mathbf{v}_{A/B} = \mathbf{v}_A - \mathbf{v}_B = (0\hat{i} + 400\hat{j}) - (300\hat{i} + 0\hat{j}) = -300\hat{i} + 400\hat{j} \text{ km/h}$. Magnitude: $\sqrt{(300^2 + 400^2)} = \sqrt{250000} = 500 \text{ km/h}$. Direction: $\arctan(400/300) = 53.1^\circ$ west of north.

Q4.

A satellite orbits Earth at 400 km altitude ($r \approx 6770 \text{ km}$ from Earth's centre, $g \approx 8.69 \text{ m/s}^2$). Find its orbital speed and period.

Answer:

$a_o = g$ at that altitude $= 8.69 \text{ m/s}^2$. $a_o = v^2/r \rightarrow v = \sqrt{gr} = \sqrt{(8.69 \times 6.77 \times 10^6)} = 7672 \text{ m/s} \approx 7.67 \text{ km/s}$. Period: $T = 2\pi r/v = 2\pi(6.77 \times 10^6)/7672 = 5543 \text{ s} \approx 92.4 \text{ min}$.

UNIT 2 — DYNAMICS & ENERGY

Chapters 5, 6, 7, 8 | The WHY of motion: forces, energy, and conservation laws

Chapter 5 — The Laws of Motion

5.1 Newton's Three Laws

Law	Statement	Mathematical form
1st — Inertia	An object at rest stays at rest; an object in motion continues at constant velocity — unless a net external force acts.	$\Sigma F = 0 \leftrightarrow a = 0$
2nd — $F = ma$	The net force on an object equals its mass times its acceleration. Both F and a are in the same direction.	$\Sigma F = ma$ (vector equation)
3rd — Action-Reaction	For every force exerted by A on B, B exerts an equal and opposite force on A. These forces act on DIFFERENT objects.	$F_{AB} = -F_{BA}$

⚠ WATCH OUT: Newton's 3rd law pairs **ALWAYS** act on different objects. They therefore **NEVER** cancel each other in a free-body diagram.

5.2 Free-Body Diagrams (FBD) — Step-by-Step

- Step 1: Isolate the object of interest. Draw it alone.
- Step 2: Identify ALL forces acting ON this object (not forces it exerts on others).
- Step 3: Draw each force as an arrow from the object's centre, in the correct direction.
- Step 4: Set up coordinate axes. Decompose forces into components.
- Step 5: Apply $\Sigma F_x = ma_x$ and $\Sigma F_y = ma_y$ separately.

Force	Symbol	Direction	When present
Weight (gravity)	$W = mg$	Downward always	Always (near Earth's surface)
Normal force	N	Perpendicular to surface, away from it	Object in contact with surface
Tension	T	Along rope, away from object	Object connected by rope/string
Friction (static)	$f_s \leq \mu_s N$	Opposing tendency to slide	No sliding occurs
Friction (kinetic)	$f_k = \mu_k N$	Opposing direction of motion	Object is sliding
Applied force	F	Direction of application	As specified

5.3 Friction — Deep Dive

Friction arises from microscopic interactions between surfaces. It always opposes relative sliding (or tendency to slide).

- **Static friction:** $f_s \leq \mu_s N$ — takes whatever value is needed to prevent sliding, up to its maximum.
- **Kinetic friction:** $f_k = \mu_k N$ — constant once sliding begins. Always: $\mu_k < \mu_s$.
- The normal force N is NOT always equal to mg . On an incline, $N = mg \cos\theta$. With an applied vertical force, N changes.

Worked Example — Block on 30° frictionless incline (find acceleration):

Forces: Weight $W = mg$ downward, Normal N perpendicular to incline

Along incline (positive down-slope): $mg \sin 30^\circ = ma \rightarrow a = g \sin 30^\circ = 9.8 \times 0.5 = 4.9 \text{ m/s}^2$

Perpendicular to incline: $N - mg \cos 30^\circ = 0 \rightarrow N = mg \cos 30^\circ$

Practice Exercises & Answers

Q1.

A 10 kg block rests on a surface with $\mu_s = 0.4$ and $\mu_k = 0.3$. What applied horizontal force is needed to (a) start it moving, and (b) keep it moving at constant speed?

Answer:

$N = mg = 10 \times 9.8 = 98 \text{ N}$. (a) $f_{s,\max} = \mu_s N = 0.4 \times 98 = 39.2 \text{ N}$ — must exceed this to start motion. (b) $f_k = \mu_k N = 0.3 \times 98 = 29.4 \text{ N}$ — must equal this to move at constant speed ($a = 0$).

Q2.

Two blocks are connected by a string over a frictionless pulley: block A (5 kg) on a frictionless table, block B (3 kg) hanging vertically. Find the acceleration of the system and the tension.

Answer:

Net force = $m_B \times g = 3 \times 9.8 = 29.4 \text{ N}$ (gravity on B drives the system). Total mass = $5 + 3 = 8 \text{ kg}$. $a = 29.4/8 = 3.675 \text{ m/s}^2$. Tension: for block A: $T = m_A \times a = 5 \times 3.675 = 18.4 \text{ N}$. Check with B: $m_B g - T = m_B a \rightarrow 29.4 - 18.4 = 11.0 = 3 \times 3.67$. ✓

Q3.

A 20 kg block sits on a 30° frictionless incline. Find the acceleration down the slope.

Answer:

Along the incline: $mg \sin\theta = ma \rightarrow a = g \sin 30^\circ = 9.8 \times 0.5 = 4.9 \text{ m/s}^2$. The normal force $N = mg \cos 30^\circ = 20 \times 9.8 \times 0.866 = 169.7 \text{ N}$ (this cancels out since there is no perpendicular acceleration).

Q4.

Why must Newton's 3rd law action-reaction pairs always act on DIFFERENT objects?

Answer:

By definition, the 3rd law says that if object A exerts a force on B, then B exerts an equal and opposite force on A. The two forces act on different objects (A and B). If they acted on the same object, they would always cancel — and nothing could ever accelerate. Action-reaction pairs should never be placed on the same free-body diagram.

Q5.

A 60 kg person stands on a scale in an elevator. The scale reads 720 N. Is the elevator accelerating? If so, in which direction and at what rate?

Answer:

Weight $W = 60 \times 9.8 = 588 \text{ N}$. Scale reads $720 \text{ N} > W$, so apparent weight is higher. $\Sigma F = N - W = ma \rightarrow 720 - 588 = 60a \rightarrow a = 132/60 = 2.2 \text{ m/s}^2$ upward. The elevator is accelerating upward (either moving up and speeding up, or moving down and slowing).

Chapter 6 — Circular Motion & Newton's Laws Applications

6.1 Centripetal Force — Concepts

For circular motion, Newton's 2nd law applied to the centripetal direction gives:

Centripetal Force

$$\Sigma F_{\text{centripetal}} = ma_c = mv^2/r = m\omega^2 r$$

This is the NET force directed INWARD toward the centre of the circle.

The centripetal force is provided by real forces: tension, gravity, friction, normal force, etc.

Scenario	What provides centripetal force
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Ball on string (horizontal circle)	Tension in string
Car rounding a flat curve	Static friction between tyres and road
Car on banked curve	Component of normal force (+ friction if needed)
Satellite orbiting Earth	Gravitational force
Ball in vertical circle (top)	Weight + tension
Ball in vertical circle (bottom)	Tension – weight

6.2 Vertical Circles — Detailed Analysis

- At the **top** of a vertical circle: both tension T and weight mg point inward.
- $\Sigma F_c = T + mg = mv^2/r \rightarrow T = mv^2/r - mg$
- Minimum speed at top ($T \rightarrow 0$): $v_{\min} = \sqrt{gr}$
- At the **bottom**: tension points inward, weight points outward.
- $\Sigma F_c = T - mg = mv^2/r \rightarrow T = mv^2/r + mg$ (tension is greatest here)

⚠ WATCH OUT: At the minimum speed at the top, gravity alone provides the centripetal force — this is why astronauts in orbit feel weightless.

6.3 Terminal Velocity

- As an object falls through air, drag force increases with speed. When drag = weight, acceleration = 0.
- Terminal velocity: $v_t = \sqrt{(2mg / D\rho A)}$ (D = drag coefficient, ρ = air density, A = cross-sectional area)

✓ **EXAM TIP: In exam problems: if asked for terminal velocity, set drag force = weight and solve for v .**

Practice Exercises & Answers

Q1.

A 0.5 kg ball on a 1.2 m string swings in a horizontal circle at 3 m/s. Find (a) the centripetal acceleration and (b) the tension in the string.

Answer:

(a) $a_c = v^2/r = 9/1.2 = 7.5 \text{ m/s}^2$ (toward centre). (b) The string is horizontal: $T = ma_c = 0.5 \times 7.5 = 3.75 \text{ N}$. (In a conical pendulum setup, weight is also involved — but here the string is assumed horizontal.)

Q2.

A 1000 kg car travels over the top of a hill of radius 40 m at 15 m/s. What is the normal force from the road on the car?

Answer:

At the top, both N and gravity act on the car; centripetal direction is downward. $mg - N = mv^2/r \rightarrow N = m(g - v^2/r) = 1000(9.8 - 225/40) = 1000(9.8 - 5.625) = 4175 \text{ N}$.

Q3.

What is the minimum speed needed at the top of a vertical loop of radius 5 m for a roller coaster car to maintain contact with the track?

Answer:

Minimum speed when $N = 0$ (track pushes with zero force). $mg = mv^2/r \rightarrow v_{\min} = \sqrt{gr} = \sqrt{(9.8 \times 5)} = \sqrt{49} = 7.0 \text{ m/s}$.

Q4.

A 70 kg skydiver reaches terminal velocity of 55 m/s. What drag force acts on them at terminal velocity?

Answer:

At terminal velocity, acceleration = 0, so drag force = weight. $F_{\text{drag}} = mg = 70 \times 9.8 = 686 \text{ N}$ upward.

Chapter 7 — Energy of a System

7.1 Work

- **Work done by a constant force:** $W = F \cdot d \cdot \cos\theta = \mathbf{F} \cdot \mathbf{d}$ (dot product)
- **Work by variable force:** $W = \int F(x)dx$ (area under F-x graph)
- **Spring force:** $F = -kx$ (Hooke's Law) $\rightarrow$ Work = $\frac{1}{2}kx^2$
- Work is POSITIVE when force component is in the direction of motion.
- Work is NEGATIVE when force component opposes motion.
- Work is ZERO when force is perpendicular to motion (e.g. centripetal force, normal force on level surface).

7.2 Kinetic Energy & Work-Energy Theorem

Work-Kinetic Energy Theorem

$$KE = \frac{1}{2}mv^2$$

$$W_{\text{net}} = \Delta KE = KE_f - KE_i$$

The net work done on an object equals the change in its kinetic energy.

✓ **EXAM TIP:** The Work-Energy Theorem relates work to change in KE. It accounts for ALL forces. This is often faster than using kinematics when time is not needed.

7.3 Potential Energy

- **Gravitational PE:** $U_g = mgy$ (y = height above reference level; reference is your choice)
- **Elastic (spring) PE:** $U_s = \frac{1}{2}kx^2$ (x = deformation from natural length)
- **Conservative force:** $W = -\Delta U$ (work by conservative force = decrease in PE)
- Examples of conservative forces: gravity, spring force, electric force.
- Examples of non-conservative forces: friction, air drag, applied forces. Cannot define PE for these.

7.4 Power

- **Average power:** $P_{\text{avg}} = W/\Delta t$ (watts, 1 W = 1 J/s)
- **Instantaneous power:** $P = dW/dt = \mathbf{F} \cdot \mathbf{v} = Fv \cos\theta$
- Conversions: 1 hp = 746 W = 550 ft·lb/s

Practice Exercises & Answers

Q1.

A force $F = 50 \text{ N}$ acts on a 10 kg box at 30° to the horizontal displacement of 8 m. Find the work done by F .

Answer:

$$W = Fd \cos\theta = 50 \times 8 \times \cos 30^\circ = 400 \times 0.866 = 346 \text{ J.}$$

Q2.

A spring with $k = 200 \text{ N/m}$ is compressed by 0.15 m. (a) What is the elastic potential energy stored? (b) If released, what is the maximum speed of a 0.5 kg block attached to it?

Answer:

(a) $U_s = \frac{1}{2}kx^2 = \frac{1}{2}(200)(0.15^2) = \frac{1}{2}(200)(0.0225) = 2.25 \text{ J}$. (b) All PE converts to KE at the natural length: $\frac{1}{2}mv^2 = 2.25 \rightarrow v = \sqrt{(2 \times 2.25/0.5)} = \sqrt{9} = 3.0 \text{ m/s}$.

Q3.

A 5 kg block moves from $v = 2 \text{ m/s}$ to $v = 6 \text{ m/s}$. How much net work was done on it?

Answer:

$$W_{\text{net}} = \Delta KE = \frac{1}{2}m(v_f^2 - v_i^2) = \frac{1}{2}(5)(36 - 4) = \frac{1}{2}(5)(32) = 80 \text{ J}.$$

Q4.

An engine delivers 15 kW of power to move a car at constant speed of 25 m/s. What is the total resistive force (drag + friction) acting on the car?

Answer:

$$\text{At constant speed, engine force} = \text{resistive force. } P = Fv \rightarrow F = P/v = 15000/25 = 600 \text{ N}.$$

Q5.

Why does the normal force on a block sliding on a flat horizontal surface do zero work?

Answer:

Work = $F \cdot d \cdot \cos\theta$. The normal force is perpendicular ($\theta = 90^\circ$) to the displacement (which is horizontal). $\cos 90^\circ = 0$, so $W = 0$. No energy is transferred by a perpendicular force.

Chapter 8 — Conservation of Energy

8.1 The Energy Conservation Equation

The most general form of energy conservation for a system:

General Energy Equation

$$\Delta E_{\text{system}} = \Sigma T$$

where T = all energy transfers across the system boundary (work, heat, etc.)

For an ISOLATED system (no external forces, no friction): $E_{\text{mech}} = \text{const}$

$$KE_i + PE_i = KE_f + PE_f$$

With FRICTION (non-conservative force):

$$KE_f + PE_f = KE_i + PE_i - f_k d$$

($f_k d$ is the thermal energy generated, not 'lost')

8.2 Problem-Solving Strategy

- Step 1: Define your system and identify the initial and final states.
- Step 2: Is the system isolated? List all non-conservative forces and their work.
- Step 3: Write the energy equation: $KE_i + PE_i + W_{nc} = KE_f + PE_f$
- Step 4: Substitute expressions. Set PE reference level where convenient (often final position or lowest point).
- Step 5: Solve for the unknown. Check units and sign.

Worked Example — 2 kg block slides down 3 m incline ($\mu_k = 0.2$, $\theta = 37^\circ$):

$$\Delta PE = -mgh = -2 \times 9.8 \times 3 \sin 37^\circ = -35.4 \text{ J (PE decreases)}$$

$$\text{Normal force } N = mg \cos 37^\circ = 2 \times 9.8 \times 0.8 = 15.7 \text{ N}$$

$$\text{Friction work } W_{\text{friction}} = -f_k \times d = -\mu_k N \times d = -0.2 \times 15.7 \times 3 = -9.4 \text{ J}$$

$$\text{Energy equation: } KE_f = KE_i + |\Delta PE| - f_k d = 0 + 35.4 - 9.4 = 26.0 \text{ J}$$

$$\text{Speed: } v = \sqrt{(2 \times 26.0/2)} = \sqrt{26.0} = 5.1 \text{ m/s}$$

⚠ WATCH OUT: When friction acts, mechanical energy is NOT conserved. But total energy (including thermal) is always conserved.

Practice Exercises & Answers

Q1.

A 2 kg ball is dropped from 10 m. What is its speed just before hitting the ground? (Use energy conservation.)

Answer:

Taking ground as PE = 0: $E_i = mgh = 2 \times 9.8 \times 10 = 196 \text{ J}$, $KE_i = 0$. $E_f = \frac{1}{2}mv^2 = 196 \text{ J} \rightarrow v = \sqrt{(2 \times 196/2)} = \sqrt{196} = 14 \text{ m/s}$.

Q2.

A 3 kg block slides down a 5 m incline ($\theta = 37^\circ$) with $\mu_k = 0.25$. Find its speed at the bottom.

Answer:

Height dropped: $h = 5\sin 37^\circ = 3.01 \text{ m}$. $\Delta PE = mgh = 3 \times 9.8 \times 3.01 = 88.4 \text{ J}$. Normal force: $N = mg \cos 37^\circ = 3 \times 9.8 \times 0.8 = 23.5 \text{ N}$. Friction work: $W_f = \mu_k Nd = 0.25 \times 23.5 \times 5 = 29.4 \text{ J}$. $KE_f = 88.4 - 29.4 = 59.0 \text{ J}$. $v = \sqrt{(2 \times 59.0/3)} = \sqrt{39.3} = 6.27 \text{ m/s}$.

Q3.

A pendulum bob (0.4 kg) is released from 20 cm above the lowest point. Find its maximum speed at the bottom.

Answer:

Energy conservation (no friction): $mgh = \frac{1}{2}mv^2 \rightarrow v = \sqrt{(2gh)} = \sqrt{(2 \times 9.8 \times 0.20)} = \sqrt{3.92} = 1.98 \text{ m/s}$.

Q4.

A machine does 4500 J of work in 1.5 minutes. What is its power output in watts and horsepower?

Answer:

$P = W/t = 4500/(1.5 \times 60) = 4500/90 = 50 \text{ W}$. In horsepower: $50/746 = 0.067 \text{ hp}$.

Q5.

Explain why a compressed spring in an isolated system has more mechanical energy than the same spring at natural length.

Answer:

Compressing a spring stores elastic potential energy $U_s = \frac{1}{2}kx^2$. In an isolated system, total mechanical energy (KE + PE) is conserved. The compressed spring has additional PE, so unless KE was reduced to provide that energy, E_{mech} is higher. When released, PE converts to KE (the block accelerates).

UNIT 3 — ELECTROSTATICS

Chapters 23, 24, 25 | Electric charges, fields, and potentials

Chapter 23 — Electric Fields

23.1 Electric Charge

- Electric charge is quantised: $Q = \pm ne$ where $e = 1.602 \times 10^{-19} \text{ C}$, $n = \text{integer}$.
- Proton: charge $+e$. Electron: charge $-e$. Neutron: charge 0 .
- Conservation of charge: net charge in an isolated system is constant.
- Conductors: charges move freely. Insulators: charges are fixed.
- Charging by induction: bring charged rod near conductor $\rightarrow$ charges redistribute $\rightarrow$ ground $\rightarrow$ remove ground $\rightarrow$ remove rod $\rightarrow$ opposite charge remains.

23.2 Coulomb's Law

Coulomb's Law

$$F = k_e |q_1||q_2| / r^2$$

$$k_e = 8.988 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2 = 1/(4\pi\epsilon_0)$$

$$\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2/(\text{N}\cdot\text{m}^2)$$

Direction: along the line joining charges (attractive if opposite, repulsive if same sign)

Magnitude form gives $|F|$; use vector diagram for direction

- Superposition: For multiple charges, find the force on the target charge from each source separately, then add VECTORIALLY.

Worked Example — Three charges on a line: $q_1 = +2\mu\text{C}$ at $x=0$, $q_2 = -3\mu\text{C}$ at $x=4\text{cm}$, find force on q_2 from q_1 :

$$F = k_e |q_1||q_2|/r^2 = (8.99 \times 10^9)(2 \times 10^{-6})(3 \times 10^{-6})/(0.04)^2 = 33.7 \text{ N}$$

Direction: $q_1(+)$ attracts $q_2(-)$, so force on q_2 is in the $-x$ direction (toward q_1)

$$F = -33.7 \text{ N } \hat{x} \text{ (in vector notation)}$$

⚠ WATCH OUT: Coulomb's law has the same mathematical form as Newton's gravity: $F \propto 1/r^2$. But electric force can be attractive OR repulsive, and is $\sim 10^{36}$ times stronger.

23.3 Electric Field

The electric field E is defined as the force per unit positive test charge. It exists whether or not a test charge is present.

Electric Field Definitions

$$E = F/q_0 \text{ (force on positive test charge } q_0, \text{ units: } \text{N/C} = \text{V/m})$$

$$\text{Point charge: } E = k_e q/r^2 \text{ (away from } +q, \text{ toward } -q)$$

$$\text{Field from multiple charges: } E_{\text{total}} = \Sigma E_i \text{ (vector sum)}$$

$$\text{Continuous distribution: } E = k_e \int dq/r^2 \times \hat{r}$$

23.4 Electric Field Lines

- Field lines originate on positive charges and terminate on negative charges.
- The tangent to a field line at any point gives the direction of E at that point.
- The density of lines (lines per unit area) is proportional to the magnitude of E .

- Field lines NEVER cross. (If they did, E would have two directions at one point — impossible.)
- Inside a conductor in electrostatic equilibrium: $E = 0$. Charge resides only on the surface.

23.5 Motion of Charged Particle in Uniform E

- A charge q in uniform E experiences constant force $F = qE$, giving constant acceleration $a = qE/m$.
- This is analogous to projectile motion — treat E -direction as the 'gravity' direction.

Worked Example — Electron enters E field ($E = 400$ N/C upward) horizontally at $v_0 = 3 \times 10^6$ m/s:

Acceleration: $a = eE/m_e = (1.6 \times 10^{-19} \times 400) / (9.11 \times 10^{-31}) = 7.02 \times 10^{13}$ m/s² upward

Use kinematic equations for x and y separately (same as projectile motion)

Practice Exercises & Answers

Q1.

Two charges $q_1 = +6 \mu\text{C}$ and $q_2 = -4 \mu\text{C}$ are separated by 0.30 m. Find the magnitude and direction of the force on q_2 .

Answer:

$F = k_e |q_1| |q_2| / r^2 = (8.99 \times 10^9)(6 \times 10^{-6})(4 \times 10^{-6}) / (0.30^2) = (8.99 \times 10^9)(24 \times 10^{-12}) / 0.09 = 2397 \times 10^{-3} / 0.09 = 2.40$ N. Direction: q_1 is positive and q_2 is negative $\rightarrow$ attractive $\rightarrow$ force on q_2 points TOWARD q_1 .

Q2.

Find the electric field at a point 0.50 m from a $+8 \mu\text{C}$ charge.

Answer:

$E = k_e q / r^2 = (8.99 \times 10^9)(8 \times 10^{-6}) / (0.50^2) = 71.92 \times 10^3 / 0.25 = 2.88 \times 10^5$ N/C, directed away from the charge.

Q3.

Three charges are on the x -axis: $+2 \mu\text{C}$ at $x = 0$, $-3 \mu\text{C}$ at $x = 0.04$ m, $+2 \mu\text{C}$ at $x = 0.08$ m. By symmetry, what is the net force on the central charge?

Answer:

The two outer charges are both $+2 \mu\text{C}$ and are equidistant (0.04 m) from the central $-3 \mu\text{C}$. Force from left charge: attractive, pointing left (toward $x = 0$). Force from right charge: attractive, pointing right (toward $x = 0.08$ m). These forces are equal in magnitude and opposite in direction $\rightarrow$ net force = 0 N.

Q4.

An electron is placed in a uniform electric field of 500 N/C pointing east. Find its acceleration (magnitude and direction).

Answer:

$F = qE = (1.6 \times 10^{-19})(500) = 8.0 \times 10^{-17}$ N. Electron is negative, so force is WESTWARD (opposite to E). $a = F/m_e = 8.0 \times 10^{-17} / 9.11 \times 10^{-31} = 8.78 \times 10^{13}$ m/s² westward.

Q5.

Sketch the electric field lines for two equal and opposite charges (electric dipole). Describe the field at the midpoint between them.

Answer:

Field lines originate at $+$ charge and terminate at $-$ charge, curving around in between. At the midpoint, contributions from both charges point in the SAME direction (from $+$ to $-$). $E_{\text{mid}} = 2 \times k_e q / r^2$ (both fields add, not cancel), directed from $+$ to $-$.

Chapter 24 — Gauss's Law

24.1 Electric Flux

- **Flux:** $\Phi_E = \int \mathbf{E} \cdot d\mathbf{A} = \int E \cos\theta \, dA$ (how much field passes through a surface)
- For uniform E through flat area A : $\Phi_E = EA \cos\theta$ (θ = angle between E and area normal)
- Flux is POSITIVE if E has a component in the outward normal direction.
- Flux is NEGATIVE if E points inward through the surface.
- Net flux through a CLOSED surface: $\Phi_E = \oint \mathbf{E} \cdot d\mathbf{A}$

24.2 Gauss's Law

Gauss's Law

$$\Phi_E = \oint \mathbf{E} \cdot d\mathbf{A} = Q_{\text{enc}}/\epsilon_0$$

The net electric flux through any closed surface equals the total enclosed charge divided by ϵ_0 .

Q_{enc} = net charge inside the Gaussian surface (positive + negative charges sum together)

Key insight: the flux through a closed surface depends ONLY on the enclosed charge — NOT on charges outside, regardless of how large or close they are.

24.3 Applying Gauss's Law — Choosing Gaussian Surfaces

Gauss's Law is ALWAYS true. But it is only useful for finding E analytically when the charge distribution has enough symmetry to let E come out of the integral.

Charge Symmetry	Gaussian Surface	Result for E
Spherical: point charge, uniform sphere, uniform shell	Concentric sphere of radius r	$E = k_e Q_{\text{enc}} / r^2$ (outward)
Cylindrical: long wire, long cylinder	Coaxial cylinder of radius r , length L	$E = \lambda / (2\pi\epsilon_0 r)$ (outward from axis) λ = charge/length
Planar: infinite flat plane	Pillbox (cylinder) straddling the plane	$E = \sigma / (2\epsilon_0)$ on each side σ = charge/area
Inside a conductor	Any surface inside conductor	$E = 0$ ($Q_{\text{enc}} = 0$, all charge on surface)
Outside a conductor	Surface just outside conductor surface	$E = \sigma / \epsilon_0$ (perpendicular to surface)

⚠ WATCH OUT: For a spherical shell of charge, $E = 0$ INSIDE the shell. This is a crucial result. For a solid sphere of uniform charge, E INSIDE increases linearly: $E = k_e Q r / R^3$.

24.4 Conductors in Electrostatic Equilibrium

1. $E = 0$ everywhere inside a conductor.
2. Any net charge resides entirely on the outer surface.
3. E at the surface is perpendicular to the surface, with magnitude σ/ϵ_0 .
4. On an irregular conductor, charge density is highest where the surface curves most (sharpest points).
5. A conductor shields its interior from external fields — Faraday cage effect.

Practice Exercises & Answers

Q1.

A solid insulating sphere of radius R carries total charge Q uniformly distributed. Find E at (a) $r > R$ and (b) $r < R$.

Answer:

(a) Outside: Gaussian sphere of radius r . $E(4\pi r^2) = Q/\epsilon_0 \rightarrow E = k_e Q/r^2$ (same as point charge). (b) Inside: charge enclosed $= Q(r/R)^3$ (scales with volume). $E(4\pi r^2) = Qr^3/(\epsilon_0 R^3) \rightarrow E = k_e Qr/R^3$ (increases linearly with r inside).

Q2.

An infinitely long straight wire carries linear charge density $\lambda = 5 \text{ nC/m}$. Find E at $r = 0.10 \text{ m}$ from the wire.

Answer:

By Gauss's Law with cylindrical surface: $E(2\pi rL) = \lambda L/\epsilon_0 \rightarrow E = \lambda/(2\pi\epsilon_0 r)$. $E = (5 \times 10^{-9})/(2\pi \times 8.85 \times 10^{-12} \times 0.10) = 5 \times 10^{-9}/5.56 \times 10^{-12} = 899 \text{ N/C}$.

Q3.

A conducting shell has inner radius 5 cm and outer radius 8 cm . A charge of $+10 \text{ nC}$ is placed at the centre. Find E at $r = 6 \text{ cm}$ (inside the shell material) and $r = 12 \text{ cm}$.

Answer:

At $r = 6 \text{ cm}$: inside the conductor $\rightarrow E = 0$. At $r = 12 \text{ cm}$: total enclosed charge $= +10 \text{ nC}$ (the inner surface of the shell gets -10 nC by induction, outer surface $+10 \text{ nC}$, net charge on shell is zero so total enclosed $= +10 \text{ nC}$). $E = k_e Q/r^2 = (8.99 \times 10^9)(10 \times 10^{-9})/(0.12^2) = 89.9/0.0144 = 6243 \text{ N/C} \approx 6.24 \text{ kN/C}$.

Q4.

Explain why the electric field inside a conductor in electrostatic equilibrium is always zero.

Answer:

If $E \neq 0$ inside a conductor, free electrons experience a net force ($F = eE$) and accelerate. This redistribution of charge continues until the field they create exactly cancels the external field, giving $E_{\text{total}} = 0$ inside. This process happens almost instantaneously. At equilibrium, no current flows and $E = 0$ everywhere inside.

Chapter 25 — Electric Potential

25.1 Electric Potential Energy & Potential

Electric potential V is potential energy per unit charge — a SCALAR quantity. Working with scalars is easier than with vectors, so V is often more convenient than E .

Key Potential Formulas

Potential difference: $\Delta V = V_B - V_A = -\int_A^B \mathbf{E} \cdot d\mathbf{s}$

Uniform field: $\Delta V = -Ed$ (d = displacement in direction of $\mathbf{E}$; V decreases in direction of $\mathbf{E}$)

Point charge: $V = k_e q/r$ ($V = 0$ at $r = \infty$; $V > 0$ near $+q$, $V < 0$ near $-q$)

Multiple charges: $V_{\text{total}} = \sum k_e q_i/r_i$ (simple SCALAR sum!)

Potential energy: $U = qV$ (energy needed to bring charge q from ∞ to that point)

Work by $\mathbf{E}$ field: $W = q(V_A - V_B) = -\Delta U$

$\mathbf{E}$ from V : $E_x = -dV/dx$, $E_y = -dV/dy$, $E_z = -dV/dz$

$\mathbf{E}$ points from HIGH V to LOW V , in the direction of steepest decrease

25.2 Equipotential Surfaces

- An equipotential surface is a surface on which $V = \text{constant}$.
- $\mathbf{E}$ field lines are ALWAYS perpendicular to equipotential surfaces.
- No work is done moving a charge along an equipotential surface.

- For a point charge: equipotentials are concentric spheres.
- For a uniform field: equipotentials are flat planes perpendicular to E.
- A conductor's surface is always an equipotential.

25.3 The Electron-Volt

- 1 eV = energy gained by one elementary charge moved through 1 V potential difference.
- 1 eV = 1.602×10^{-19} J.
- Very useful for atomic/sub-atomic scale energies.

Worked Example — Find speed of proton accelerated through $\Delta V = 1000$ V:

Energy gained: $\Delta KE = q\Delta V = (1.6 \times 10^{-19})(1000) = 1.6 \times 10^{-16}$ J

$\frac{1}{2}m_p v^2 = 1.6 \times 10^{-16} \rightarrow v = \sqrt{(2 \times 1.6 \times 10^{-16} / 1.67 \times 10^{-27})} = 4.38 \times 10^5$ m/s

⚠ WATCH OUT: Positive charges move spontaneously from high V to low V. Negative charges (electrons) move spontaneously from low V to high V.

Practice Exercises & Answers

Q1.

Find the electric potential at 0.30 m from a $+5 \mu\text{C}$ point charge.

Answer:

$$V = k_e q / r = (8.99 \times 10^9)(5 \times 10^{-6}) / 0.30 = 44950 / 0.30 = 149\,833 \text{ V} \approx 150 \text{ kV.}$$

Q2.

Two charges: $+3 \mu\text{C}$ at origin and $-3 \mu\text{C}$ at $x = 0.06$ m. Find the potential at the midpoint $x = 0.03$ m.

Answer:

$V = k_e q_1 / r_1 + k_e q_2 / r_2 = k_e (+3 \times 10^{-6}) / 0.03 + k_e (-3 \times 10^{-6}) / 0.03$. Both terms are equal and opposite $\rightarrow V = 0$. (Potential is zero at the midpoint of an electric dipole, but $E \neq 0$ there.)

Q3.

A proton is accelerated from rest through a potential difference of 2000 V. Find its final kinetic energy and speed.

Answer:

$$KE = q\Delta V = (1.6 \times 10^{-19})(2000) = 3.2 \times 10^{-16} \text{ J. } \frac{1}{2}m_p v^2 = 3.2 \times 10^{-16} \rightarrow v = \sqrt{(2 \times 3.2 \times 10^{-16} / 1.67 \times 10^{-27})} = \sqrt{(3.83 \times 10^{11})} = 6.19 \times 10^5 \text{ m/s.}$$

Q4.

In a uniform electric field $E = 800$ N/C pointing in the $+x$ direction, find the potential difference $V_A - V_B$ where B is 0.05 m to the right of A.

Answer:

$\Delta V = V_B - V_A = -Ed = -800 \times 0.05 = -40$ V. Therefore $V_A - V_B = +40$ V. (Potential decreases in the direction E points, so A is at higher potential than B.)

Q5.

Why is it easier to use electric potential (a scalar) instead of electric field (a vector) when calculating work done on charges?

Answer:

Work done by E field: $W = q(V_A - V_B)$. This requires only the scalar values of potential at two points — one subtraction and one multiplication. Using E would require integrating the vector field along the path, accounting for direction at every point. The scalar nature of V makes superposition trivial (just add numbers, not vectors).

UNIT 4 — CAPACITORS & DC CIRCUITS

Chapters 26, 27, 28 | Storing charge, moving charge, and analysing circuits

Chapter 26 — Capacitance & Dielectrics

26.1 Capacitance — Concept & Definition

A capacitor consists of two conductors separated by an insulating gap. When connected to a battery, opposite charges $\pm Q$ build up on the plates, creating a potential difference $V = Q/C$ between them.

Capacitance Formulas

Definition: $C = Q/V$ (farads, $F = C/V$)

Parallel plate (air gap): $C = \epsilon_0 A/d$ ($\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$, A = plate area, d = separation)

Parallel plate (dielectric): $C = \kappa \epsilon_0 A/d$ (κ = dielectric constant ≥ 1)

Cylindrical: $C = 2\pi\epsilon_0 L / \ln(b/a)$ (inner radius a , outer radius b , length L)

Spherical: $C = 4\pi\epsilon_0 \times (ab/(b-a))$ (inner radius a , outer radius b)

26.2 Capacitors in Series and Parallel

SERIES CAPACITORS:

$$1/C_{eq} = 1/C_1 + 1/C_2 + 1/C_3$$

Same charge Q on each capacitor

$$\text{Voltages add: } V = V_1 + V_2 + V_3$$

C_{eq} is LESS than smallest C

PARALLEL CAPACITORS:

$$C_{eq} = C_1 + C_2 + C_3$$

Same voltage V across each

$$\text{Charges add: } Q = Q_1 + Q_2 + Q_3$$

C_{eq} is MORE than largest C

Worked Example — $3 \mu\text{F}$ and $6 \mu\text{F}$ in series, then that combination in parallel with $4 \mu\text{F}$:

Series: $1/C_s = 1/3 + 1/6 = 3/6 \rightarrow C_s = 2 \mu\text{F}$

Parallel with $4 \mu\text{F}$: $C_{eq} = 2 + 4 = 6 \mu\text{F}$

26.3 Energy Stored in a Capacitor

Energy in a Capacitor

$$U = Q^2/(2C) = \frac{1}{2}CV^2 = \frac{1}{2}QV$$

All three forms are equivalent; use whichever matches the known quantities.

Energy density in E field: $u = \frac{1}{2}\epsilon_0 E^2 \text{ (J/m}^3\text{)}$ [or $u = \frac{1}{2}\kappa\epsilon_0 E^2$ with dielectric]

26.4 Dielectrics — Detailed Effects

Quantity	Effect of inserting dielectric ($\kappa > 1$)	Disconnected (Q fixed)	Connected to battery (V fixed)
Capacitance C	Increases by factor κ	κC_0	κC_0
Charge Q	Changes	Stays Q_0	Increases to κQ_0
Voltage V	Changes	Decreases to V_0/κ	Stays V_0
Electric field E	Decreases	E_0/κ	E_0 (same as without dielectric — wrong! it's E_0)
Energy stored U	Changes	Decreases to U_0/κ	Increases to κU_0

△ WATCH OUT: Whether Q or V is fixed depends entirely on whether the capacitor is **DISCONNECTED** from or **CONNECTED TO** the battery when the dielectric is inserted. This distinction is critical.

Practice Exercises & Answers

Q1.

A parallel-plate capacitor has plates of area 0.02 m^2 separated by 0.5 mm . Find (a) its capacitance in air and (b) with a dielectric of $\kappa = 4.5$.

Answer:

(a) $C = \epsilon_0 A/d = (8.85 \times 10^{-12})(0.02)/(0.5 \times 10^{-3}) = 177 \times 10^{-12} / 0.5 \times 10^{-3} = 354 \text{ pF}$. (b) $C' = \kappa C = 4.5 \times 354 \text{ pF} = 1593 \text{ pF} \approx 1.59 \text{ nF}$.

Q2.

Three capacitors $C_1 = 2 \text{ }\mu\text{F}$, $C_2 = 3 \text{ }\mu\text{F}$, $C_3 = 6 \text{ }\mu\text{F}$ are connected in parallel to a 12 V battery. Find the equivalent capacitance, total charge, and energy stored.

Answer:

$C_{\text{eq}} = 2 + 3 + 6 = 11 \text{ }\mu\text{F}$. $Q_{\text{total}} = C_{\text{eq}} \times V = 11 \times 10^{-6} \times 12 = 132 \text{ }\mu\text{C}$. $U = \frac{1}{2} C_{\text{eq}} V^2 = \frac{1}{2} (11 \times 10^{-6}) (144) = 792 \text{ }\mu\text{J}$.

Q3.

The same three capacitors are connected in series to 12 V . Find C_{eq} , the charge on each, and voltage across each.

Answer:

$1/C_{\text{eq}} = 1/2 + 1/3 + 1/6 = 3/6 + 2/6 + 1/6 = 6/6 = 1 \rightarrow C_{\text{eq}} = 1 \text{ }\mu\text{F}$. $Q = C_{\text{eq}} \times V = 1 \times 12 = 12 \text{ }\mu\text{C}$ (same on each in series). $V_1 = 12/2 = 6 \text{ V}$, $V_2 = 12/3 = 4 \text{ V}$, $V_3 = 12/6 = 2 \text{ V}$. Check: $6+4+2 = 12 \text{ V}$. ✓

Q4.

A $50 \text{ }\mu\text{F}$ capacitor is charged to 200 V then disconnected. A dielectric with $\kappa = 2$ is inserted. Find the new voltage and energy. What happened to the 'missing' energy?

Answer:

Initial: $Q = CV = 50 \times 10^{-6} \times 200 = 10^{-2} \text{ C} = 10 \text{ mC}$. $U_i = \frac{1}{2} CV^2 = \frac{1}{2} (50 \times 10^{-6}) (40000) = 1 \text{ J}$.
Disconnected $\rightarrow Q$ fixed at 10 mC . New $C = \kappa C_0 = 100 \text{ }\mu\text{F}$. New $V = Q/C' = 10^{-2} / 100 \times 10^{-6} = 100 \text{ V}$.
New $U = Q^2/2C' = (10^{-2})^2 / (2 \times 10^{-4}) = 10^{-4} / 2 \times 10^{-4} = 0.5 \text{ J}$. The 'missing' 0.5 J was used as work done to pull the dielectric into the field (or dissipated).

Chapter 27 — Current & Resistance

27.1 Electric Current

- **Current I :** rate of charge flow. $I = dQ/dt$ (amperes, $A = C/s$)
- Conventional current direction: direction positive charges would flow (opposite to electron drift).
- **Current density J :** $J = I/A$ (A/m^2), where A is cross-sectional area.
- **Drift velocity v_d :** $I = nqv_d A$ (n = carrier number density, q = carrier charge)
- Drift speed is very slow ($\sim 10^{-4} \text{ m/s}$), but the electric signal travels near light speed.

27.2 Resistance & Ohm's Law

Resistance & Resistivity

Ohm's Law: $V = IR$ (R in ohms, $\Omega = V/A$)

Resistance from geometry: $R = \rho L/A$

(ρ = resistivity in $\Omega \cdot \text{m}$, L = length, A = cross-sectional area)

Temperature dependence: $\rho = \rho_0[1 + \alpha(T - T_0)]$ (α = temperature coefficient)

R increases with T for metals ($\alpha > 0$); decreases for semiconductors ($\alpha < 0$)

Material	Resistivity ρ ($\Omega \cdot \text{m}$) at 20°C	Type
Silver	1.59×10^{-8}	Conductor
Copper	1.70×10^{-8}	Conductor
Aluminum	2.82×10^{-8}	Conductor
Silicon	640	Semiconductor
Glass	$10^{10} - 10^{14}$	Insulator

27.3 Electrical Power

Power Dissipated in a Resistor

$$P = IV = I^2R = V^2/R$$

All three forms are equivalent. Use whichever has two known quantities.

Energy dissipated: $E = Pt = I^2Rt = V^2t/R$ (joules)

This energy is converted to thermal energy (heat) in the resistor.

- **Superconductors:** below critical temperature T_c , resistance = 0 exactly. No energy is dissipated. Current flows indefinitely without a voltage source.

✓ **EXAM TIP:** Remember $P = V^2/R$: a device uses **MORE power at higher voltage (for fixed R)**, but if devices are in series (same I), use $P = I^2R$ — higher R uses more power.

Practice Exercises & Answers

Q1.

A copper wire ($\rho = 1.7 \times 10^{-8} \Omega \cdot \text{m}$) has length 2 m and diameter 1 mm. Find its resistance.

Answer:

$$A = \pi(d/2)^2 = \pi(0.5 \times 10^{-3})^2 = 7.85 \times 10^{-7} \text{ m}^2. \quad R = \rho L/A = (1.7 \times 10^{-8})(2)/(7.85 \times 10^{-7}) = 3.4 \times 10^{-8}/7.85 \times 10^{-7} = 0.0433 \Omega \approx 43.3 \text{ m}\Omega.$$

Q2.

A 60 W light bulb operates at 120 V. Find (a) the current drawn and (b) the resistance of the filament.

Answer:

$$(a) P = IV \rightarrow I = P/V = 60/120 = 0.5 \text{ A. (b) } R = V/I = 120/0.5 = 240 \Omega. \text{ (Or: } R = V^2/P = 14400/60 = 240 \Omega.)$$

Q3.

A resistor has resistance 100 Ω at 20°C. At 100°C its resistance is 140 Ω . Find the temperature coefficient α .

Answer:

$$R = R_0[1 + \alpha(T - T_0)] \rightarrow 140 = 100[1 + \alpha(80)] \rightarrow 1.4 = 1 + 80\alpha \rightarrow \alpha = 0.4/80 = 5.0 \times 10^{-3} / ^\circ\text{C}.$$

Q4.

Calculate the electrical energy consumed by a 1000 W microwave used for 5 minutes, in joules and in kWh.

Answer:

$$E = Pt = 1000 \times (5 \times 60) = 300\,000 \text{ J} = 300 \text{ kJ. In kWh: } E = 1 \text{ kW} \times (5/60) \text{ h} = 0.0833 \text{ kWh.}$$

Q5.

Two resistors $R_1 = 4\ \Omega$ and $R_2 = 12\ \Omega$ carry the same current of 3 A (in series). Compare the power dissipated in each.

Answer:

$P_1 = I^2 R_1 = 9 \times 4 = 36\text{ W}$. $P_2 = I^2 R_2 = 9 \times 12 = 108\text{ W}$. The larger resistor dissipates 3× more power ($P \propto R$ when I is fixed in series).

Chapter 28 — Direct-Current Circuits

28.1 EMF and Real Batteries

- **EMF (electromotive force) $\mathcal{E}$:** energy per unit charge provided by the battery (not really a force).
- **Internal resistance r :** every real battery has internal resistance. Current flows through it, causing a voltage drop.

Real Battery

Terminal voltage: $V_{\text{terminal}} = \mathcal{E} - Ir$

(Less than EMF when current flows out of battery; more than EMF when charging)

Power delivered to external circuit: $P_{\text{ext}} = I(\mathcal{E} - Ir) = IV_{\text{terminal}}$

Power dissipated internally: $P_{\text{int}} = I^2 r$

28.2 Resistors in Series and Parallel

SERIES RESISTORS:

$$R_{\text{eq}} = R_1 + R_2 + R_3 + \dots$$

Same current I through each

Voltages add: $V = V_1 + V_2 + V_3$

$R_{\text{eq}} > \text{any individual } R$

PARALLEL RESISTORS:

$$1/R_{\text{eq}} = 1/R_1 + 1/R_2 + \dots$$

Same voltage V across each

Currents add: $I = I_1 + I_2 + \dots$

$R_{\text{eq}} < \text{any individual } R$

- Special case — two resistors in parallel: $R_{\text{eq}} = R_1 R_2 / (R_1 + R_2)$

⚠ WATCH OUT: In parallel, current divides *INVERSELY* proportional to resistance. More current flows through the *SMALLER* resistor.

28.3 Kirchhoff's Rules — Systematic Circuit Analysis

For circuits that cannot be reduced by series/parallel combinations (e.g. bridge circuits, multi-loop circuits), use Kirchhoff's Rules.

Kirchhoff's Rules

JUNCTION RULE (KCL — charge conservation):

$$\Sigma I_{\text{in}} = \Sigma I_{\text{out}} \text{ at every node}$$

LOOP RULE (KVL — energy conservation):

$$\Sigma \Delta V = 0 \text{ around any closed loop}$$

SIGN CONVENTIONS for the loop rule:

Resistor: traverse in direction of $I \rightarrow \Delta V = -IR$ (voltage drops)

Resistor: traverse opposite to $I \rightarrow \Delta V = +IR$ (voltage rises)

Battery: traverse – to + (through battery) $\rightarrow \Delta V = +\mathcal{E}$

Battery: traverse + to – (through battery) $\rightarrow \Delta V = -\mathcal{E}$

28.4 Kirchhoff's Rules — Step-by-Step Method

- Step 1: Label all unknown currents with arrows. If your guess is wrong, the answer will be negative (which is fine).
- Step 2: Apply KCL at every independent junction. If there are n junctions, write $n-1$ independent equations.
- Step 3: Identify all independent loops. Draw loop direction arrows (clockwise or counterclockwise — your choice).
- Step 4: Apply KVL to enough independent loops to complete the system of equations.
- Step 5: Solve the simultaneous equations. Negative current means it flows opposite to your assumed direction.

Quick Kirchhoff example — single loop: 12V battery ($r=1\Omega$), then 5Ω and 3Ω in series:

$$\text{KVL: } +12 - I(1) - I(5) - I(3) = 0$$

$$12 = 9I \rightarrow I = 12/9 = 1.33 \text{ A}$$

$$V_{5\Omega} = 1.33 \times 5 = 6.67 \text{ V}, V_{3\Omega} = 1.33 \times 3 = 4.0 \text{ V}, V_{\text{internal}} = 1.33 \times 1 = 1.33 \text{ V}$$

28.5 RC Circuits — Charging and Discharging

When a capacitor charges through a resistor, the current and voltage change exponentially, NOT instantaneously.

RC Circuit Equations

Time constant: $\tau = RC$ (units: seconds, $\Omega \times F = s$)

CHARGING (initially uncharged, switch closed at $t=0$):

$$q(t) = C\varepsilon(1 - e^{-t/RC}) = Q_{\text{max}}(1 - e^{-t/\tau})$$

$$V_C(t) = \varepsilon(1 - e^{-t/\tau})$$

$$I(t) = (\varepsilon/R)e^{-t/\tau} \text{ (starts at max, decays to zero)}$$

DISCHARGING (initially charged to Q_0 , battery disconnected):

$$q(t) = Q_0 e^{-t/\tau}$$

$$V_C(t) = V_0 e^{-t/\tau}$$

$$I(t) = (V_0/R)e^{-t/\tau} \text{ (starts at max, decays to zero)}$$

Time elapsed	Charge reached (charging)	Charge remaining (discharging)
$t = \tau = RC$	63.2% of Q_{max}	36.8% of Q_0
$t = 2\tau$	86.5% of Q_{max}	13.5% of Q_0
$t = 3\tau$	95.0% of Q_{max}	5.0% of Q_0
$t = 5\tau$	99.3% of Q_{max}	0.7% of Q_0 ($\approx$ fully charged/discharged)
$t = 0$	0%	100%

- At $t = 0$ (capacitor uncharged): capacitor acts like a SHORT CIRCUIT (wire). $V_C = 0$.
- At $t = \infty$ (fully charged): capacitor acts like an OPEN CIRCUIT. $I = 0$. All voltage across capacitor.

⚠ WATCH OUT: Never assume a capacitor charges instantaneously. The time constant $\tau = RC$ always governs the rate.

Practice Exercises & Answers

Q1.

A battery ($\varepsilon = 9 \text{ V}$, $r = 0.5 \Omega$) is connected to an external $R = 8.5 \Omega$. Find (a) the current, (b) the terminal voltage, and (c) power to external R .

Answer:

(a) $I = \varepsilon / (R + r) = 9 / (8.5 + 0.5) = 9 / 9 = 1.0 \text{ A}$. (b) $V_{\text{term}} = \varepsilon - Ir = 9 - 1.0 \times 0.5 = 8.5 \text{ V}$. (Or $V = IR = 1.0 \times 8.5 = 8.5 \text{ V}$. ✓) (c) $P_{\text{ext}} = I^2 R = 1^2 \times 8.5 = 8.5 \text{ W}$.

Q2.

Three resistors $R_1 = 6 \Omega$, $R_2 = 3 \Omega$ (in parallel), and $R_3 = 4 \Omega$ in series with the combination, connected to 18 V . Find the current through each resistor.

Answer:

Parallel pair: $1/R_p = 1/6 + 1/3 = 1/6 + 2/6 = 3/6 \rightarrow R_p = 2 \Omega$. Total: $R_{\text{eq}} = R_p + R_3 = 2 + 4 = 6 \Omega$. $I_{\text{total}} = 18/6 = 3 \text{ A}$ (through R_3). V across parallel pair $= I \times R_p = 3 \times 2 = 6 \text{ V}$. $I_1 = 6/6 = 1 \text{ A}$, $I_2 = 6/3 = 2 \text{ A}$. Check: $1 + 2 = 3 \text{ A}$. ✓

Q3.

Apply Kirchhoff's rules to a two-loop circuit: loop 1 has $\varepsilon_1 = 12 \text{ V}$ with $R_1 = 2 \Omega$; loop 2 has $\varepsilon_2 = 8 \text{ V}$ with $R_2 = 3 \Omega$; shared middle branch has $R_3 = 1 \Omega$. Find the three branch currents (assume I_1 flows in loop 1 clockwise, I_2 in loop 2 clockwise, $I_3 = I_1 - I_2$ in the middle branch downward).

Answer:

KCL at top junction: $I_3 = I_1 - I_2$. KVL loop 1 (clockwise): $12 - 2I_1 - 1(I_1 - I_2) = 0 \rightarrow 12 = 3I_1 - I_2 \dots (i)$
KVL loop 2 (clockwise): $8 - 3I_2 + 1(I_1 - I_2) = 0 \rightarrow 8 = -I_1 + 4I_2 \dots (ii)$ From (i): $I_2 = 3I_1 - 12$. Sub into (ii): $8 = -I_1 + 4(3I_1 - 12) = 11I_1 - 48 \rightarrow I_1 = 56/11 \approx 5.09 \text{ A}$. $I_2 = 3(5.09) - 12 = 3.27 \text{ A}$. $I_3 = 5.09 - 3.27 = 1.82 \text{ A}$.

Q4.

A $100 \mu\text{F}$ capacitor is charged through a $10 \text{ k}\Omega$ resistor from a 24 V supply. Find (a) the time constant, (b) the charge after one time constant, and (c) the voltage across the capacitor after 2τ .

Answer:

(a) $\tau = RC = 10 \times 10^3 \times 100 \times 10^{-6} = 1.0 \text{ s}$. (b) $Q_{\text{max}} = C\varepsilon = 100 \times 10^{-6} \times 24 = 2.4 \text{ mC}$. After 1τ : $q = Q_{\text{max}}(1 - e^{-1}) = 2.4 \times 0.632 = 1.517 \text{ mC}$. (c) $V_C(2\tau) = \varepsilon(1 - e^{-2}) = 24(1 - 0.135) = 24 \times 0.865 = 20.76 \text{ V}$.

Q5.

At $t = 0$, a capacitor in an RC circuit is fully charged to $V_0 = 50 \text{ V}$. The capacitor then discharges through $R = 5 \text{ k}\Omega$. After 3 ms the voltage is 25 V . Find C .

Answer:

$V(t) = V_0 e^{(-t/RC)} \rightarrow 25 = 50 e^{(-0.003/RC)} \rightarrow 0.5 = e^{(-0.003/RC)}$. Take $\ln$: $\ln(0.5) = -0.003/RC \rightarrow -0.693 = -0.003/RC \rightarrow RC = 0.003/0.693 = 4.33 \times 10^{-3} \text{ s}$. $C = RC/R = 4.33 \times 10^{-3}/5000 = 8.66 \times 10^{-7} \text{ F} \approx 0.866 \mu\text{F}$.

COMPREHENSIVE FORMULA CHEAT SHEET

All key equations, constants, and unit conversions from Units 1-4

Unit 1 — Kinematics & Vectors

Category	Formula	Notes / Conditions
Displacement	$\Delta x = x_f - x_i$	Vector — can be negative
Avg velocity	$v_{avg} = \Delta x / \Delta t$	Same direction as Δx
Instantaneous v	$v = dx/dt$	Slope of x-t graph
Acceleration	$a = dv/dt = d^2x/dt^2$	Slope of v-t graph
Kinematic #1	$v = v_0 + at$	No x
Kinematic #2	$x = x_0 + v_0t + \frac{1}{2}at^2$	No v
Kinematic #3	$v^2 = v_0^2 + 2a(x - x_0)$	No t
Kinematic #4	$x = x_0 + \frac{1}{2}(v_0 + v)t$	No a
Free fall	$a = -g = -9.80 \text{ m/s}^2$	Upward positive convention
Vector components	$A_x = A \cos \theta, A_y = A \sin \theta$	θ from +x axis
Vector magnitude	$A = \sqrt{A_x^2 + A_y^2}$	Always positive
Dot product	$A \cdot B = AB \cos \theta = A_x B_x + A_y B_y + A_z B_z$	Scalar result
Projectile x	$x = v_0 \cos \theta \cdot t$	Constant velocity, no gravity
Projectile y	$y = v_0 \sin \theta \cdot t - \frac{1}{2}gt^2$	Free fall
Range	$R = v_0^2 \sin(2\theta) / g$	Level ground, max at 45°
Centripetal acc.	$a_c = v^2/r = \omega^2 r$	Directed inward
Relative velocity	$v_{AB} = v_{AE} - v_{BE}$	Vector subtraction

Unit 2 — Dynamics & Energy

Category	Formula	Notes / Conditions
Newton 2nd	$\Sigma F = ma$ (vector)	Apply per axis: $\Sigma F_x = max, \Sigma F_y = may$
Weight	$W = mg$	Always downward, $g = 9.80 \text{ m/s}^2$
Normal force	N (varies)	Perpendicular to surface
Static friction	$f_s \leq \mu_s N$	Prevents sliding; takes any value up to max
Kinetic friction	$f_k = \mu_k N$	Opposes sliding; $\mu_k < \mu_s$
Centripetal force	$\Sigma F_c = mv^2/r$	Net force toward center
Work (constant F)	$W = Fd \cos \theta$	θ between F and d; $W < 0$ if opposing
Work-Energy Thm	$W_{net} = \Delta KE$	Applies to all forces combined
Kinetic Energy	$KE = \frac{1}{2}mv^2$	Always ≥ 0
Grav. PE	$U_g = mgy$	Reference at $y = 0$ (your choice)
Spring PE	$U_s = \frac{1}{2}kx^2$	x = deformation from natural length
Energy conserv.	$KE_i + PE_i = KE_f + PE_f$	Isolated system, no friction
With friction	$\Delta E_{mech} = -f_k d$	Energy $\rightarrow$ thermal; not 'lost'

Power (avg)	$P = W/\Delta t = Fv \cos\theta$	Watts ($W = J/s$)
Drag (high speed)	$R = \frac{1}{2}D\rho Av^2$	D = drag coeff., ρ = fluid density
Terminal velocity	$v_t = \sqrt{(2mg/D\rho A)}$	When drag = weight

Unit 3 — Electrostatics

Category	Formula	Notes / Conditions
Coulomb's Law	$F = k_e q_1 q_2 /r^2$	$k_e = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$
Permittivity	$\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/(\text{N}\cdot\text{m}^2)$	$k_e = 1/(4\pi\epsilon_0)$
Electric field	$E = F/q_0 = k_e q /r^2$	Direction away from +, toward –
E superposition	$E_{\text{total}} = \sum E_i$ (vectors)	Add all contributions vectorially
E continuous	$E = k_e \int dq/r^2 \hat{r}$	Integrate over charge distribution
Electric flux	$\Phi E = EA \cos\theta = \oint E \cdot dA$	θ between E and area normal
Gauss's Law	$\Phi E = Q_{\text{enc}}/\epsilon_0$	Any closed surface
E: sphere outside	$E = k_e Q/r^2$	Same as point charge
E: long wire	$E = \lambda/(2\pi\epsilon_0 r)$	λ = charge per length
E: infinite plane	$E = \sigma/(2\epsilon_0)$	σ = surface charge density
E: conductor	$E = 0$ inside; $E = \sigma/\epsilon_0$ outside	Electrostatic equilibrium
Potential diff.	$\Delta V = -\int E \cdot ds$	V decreases in direction of E
V uniform field	$\Delta V = -Ed$	d parallel to E
V point charge	$V = k_e q/r$	$V = 0$ at $r = \infty$
V superposition	$V = \sum k_e q_i/r_i$	SCALAR sum (easier than E)
PE of charge	$U = qV$	Potential energy at that location
E from V	$E = -dV/dx$ (component)	E points from high V to low V
Work by E	$W = q(V_A - V_B) = -\Delta U$	Positive work if charge moves to lower PE
Electron-volt	$1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$	Energy of e^- through 1 V

Unit 4 — Capacitors & DC Circuits

Category	Formula	Notes / Conditions
Capacitance	$C = Q/V$	Farads ($F = C/V$)
Parallel plate	$C = \epsilon_0 A/d$ or $\kappa\epsilon_0 A/d$	κ = dielectric constant ≥ 1
Series caps	$1/C_{\text{eq}} = \sum 1/C_i$	Same Q , different V
Parallel caps	$C_{\text{eq}} = \sum C_i$	Same V , different Q
Energy in cap.	$U = Q^2/2C = \frac{1}{2}CV^2 = \frac{1}{2}QV$	Use form with known quantities
E field energy	$u = \frac{1}{2}\epsilon_0 E^2$	Energy density (J/m^3)
Dielectric effect	$C \rightarrow \kappa C$, $E \rightarrow E/\kappa$, U changes	Depends on whether Q or V is fixed
Current	$I = dQ/dt$	Amperes ($A = C/s$)
Current density	$J = I/A = nqv_d$	n = carrier density, v_d = drift speed
Ohm's Law	$V = IR$	Ohms ($\Omega = V/A$)
Resistivity	$R = \rho L/A$	ρ in $\Omega\cdot m$
Temp. coeff.	$\rho = \rho_0[1 + \alpha(T - T_0)]$	Metals: $\alpha > 0$ (R increases with T)

Power	$P = IV = I^2R = V^2/R$	Watts. Use form with two knowns.
Resistors series	$R_{eq} = \sum R_i$	Same I, voltages add
Resistors parallel	$1/R_{eq} = \sum 1/R_i$	Same V, currents add
Two parallel	$R_{eq} = R_1 R_2 / (R_1 + R_2)$	Shortcut for two resistors
EMF/terminal V	$V_{term} = \mathcal{E} - Ir$	r = internal resistance
KCL (junction)	$\sum I_{in} = \sum I_{out}$	Charge conservation
KVL (loop)	$\sum \Delta V = 0$	Energy conservation
RC time constant	$\tau = RC$	Seconds
RC charging	$q = C\mathcal{E}(1 - e^{-(t/\tau)}), I = (\mathcal{E}/R)e^{-(t/\tau)}$	Initially uncharged
RC discharging	$q = Q_0 e^{-(t/\tau)}, I = (V_0/R)e^{-(t/\tau)}$	Initially charged to Q_0
At $t=0$ (uncharge)	Cap. = short circuit (wire)	$V_C = 0, I = \mathcal{E}/R$
At $t=\infty$	Cap. = open circuit	$I = 0, V_C = \mathcal{E}$

Physical Constants & Conversions

Constant	Symbol	Value
Elementary charge	e	$1.602 \times 10^{-19} \text{ C}$
Electron mass	m_e	$9.109 \times 10^{-31} \text{ kg}$
Proton mass	m_p	$1.673 \times 10^{-27} \text{ kg}$
Gravitational acc. (surface)	g	9.80 m/s^2
Coulomb's constant	k_e	$8.988 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$
Permittivity of free space	ϵ_0	$8.854 \times 10^{-12} \text{ C}^2/(\text{N}\cdot\text{m}^2)$
Speed of light in vacuum	c	$2.998 \times 10^8 \text{ m/s}$
Gravitational constant	G	$6.674 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2$
1 eV	—	$1.602 \times 10^{-19} \text{ J}$
1 hp (horsepower)	—	746 W
Atmospheric pressure	P_0	$1.013 \times 10^5 \text{ Pa}$

EXAM STRATEGY: (1) Read the problem twice — identify what is GIVEN and what is ASKED. (2) Draw a diagram or FBD. (3) Select the correct physical law or formula. (4) Apply it symbolically first, then substitute numbers. (5) Check: are units correct? Is the magnitude reasonable? Does the direction make physical sense?